Effect of Wiener Speech Enhancement on Speaker Identity Preservation Under Noise

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1. Introduction

1.1 Background

Speech enhancement aims to reduce background noise and improve the quality of speech recordings. Wiener filtering is a classical method that can reduce noise in the time-frequency domain [1]. Although this method can improve signal quality, the enhancement process may also change useful speech characteristics.

Speaker verification determines whether different recordings belong to the same speaker. Modern systems such as ECAPA-TDNN extract speaker embeddings to represent speaker identity [2].

These technologies are widely relevant to practical applications such as voice authentication, access control, and remote communication, where background noise may affect system reliability. Speech enhancement can help improve audio quality in these environments, but changes to speaker-related features may also influence verification accuracy.

Therefore, improvements in speech quality do not necessarily lead to better speaker verification results. Understanding this relationship is important for developing reliable speech processing systems in noisy environments.

1.2 Research Question and Objectives

In this project, I will investigate the following research question:

Does Wiener speech enhancement improve or damage speaker identity information under different noise conditions?

And a related question will also be discussed which is whether signal-to-noise ratio (SNR) improves while speaker verification performance becomes worse.

In this experiment I used speech recordings from LibriSpeech [3] and environmental noise from DEMAND [4]. Ten speakers, three noise environments, and four input SNR levels (−5, 0, +5, and +10 dB) are included.

And a Wiener filter was applied to the noisy recordings. Clean, noisy, and enhanced speech were compared by using SNR improvement, Mel-filterbank distortion, speaker embedding drift, and equal error rate (EER).

1.3 Project Contributions

The main contribution of this project is an experimental evaluation of how classical Wiener filtering affects speaker identity information.

In this project, a MATLAB-based speech enhancement pipeline was implemented to generate noisy recordings and apply Wiener filtering. A Python-based evaluation pipeline was developed using a pretrained ECAPA-TDNN model to extract speaker embeddings and calculate verification scores.

By comparing signal quality and speaker-related metrics across different noise conditions, this project shows whether speech enhancement improves or damages speaker identity information.

2. Literature Review

2.1 Wiener Filtering for Speech Enhancement

Nowadays, Various speech enhancement methods have been developed to reduce background noise while minimizing speech distortion. Among them, classical spectral estimation methods provide a foundation for understanding the trade-off between noise reduction and speech preservation.

Lim and Oppenheim [1] discussed classical speech enhancement methods based on spectral estimation. Wiener filtering is one such approach, aiming to minimize the mean squared error between the estimated and original speech signals. Its performance depends on the accuracy of speech and noise power estimates, making it sensitive to changes in noise conditions.

Ephraim and Malah [5] introduced a minimum mean-square error (MMSE) short-time spectral amplitude estimator. Unlike conventional Wiener filtering, which minimizes the error of the complex spectral estimate, their approach focuses on estimating spectral amplitude. This difference shows that speech enhancement methods can use different estimation objectives, which may lead to different noise reduction and speech distortion characteristics.

Although more advanced estimators are available, Wiener filtering provides a relatively simple and well-established baseline. However, minimizing signal estimation error does not

directly guarantee that speaker-specific characteristics will be preserved. This distinction is important because speaker verification depends on identity-related features rather than signal quality alone. Therefore, Wiener filtering provides a suitable starting point for investigating the relationship between noise reduction and speaker identity preservation.

2.2 Speaker Verification and ECAPA-TDNN

With the development of deep learning, speaker verification systems have increasingly used neural networks to extract speaker-related features from speech signals. These features are speaker embeddings, allowing different recordings to be compared based on speaker identity.

Snyder et al. [6] introduced x-vectors, which use deep neural networks to generate fixed-dimensional speaker representations. Their research showed that data augmentation with noise and reverberation can improve speaker recognition robustness.

Building on earlier TDNN-based approaches, Desplanques et al. [2] proposed ECAPA-TDNN, which combines channel attention, multi-layer feature aggregation, and attentive statistics pooling. These improvements allow the model to capture speaker-related information more effectively and achieve better verification performance than earlier TDNN-based systems.

Although these models have improved speaker verification performance, their extracted embeddings may still be affected by noise and speech processing. Therefore, evaluating changes in speaker embeddings is important when examining whether speech enhancement preserves speaker identity information.

2.3 Research Gap and Evaluation Approach

Although speech enhancement and speaker verification have been widely used, their relationship is still an important research topic. Traditional speech enhancement methods mainly focus on improving signal quality, while speaker verification systems aim to identify speakers' characteristics. But better speech quality may not always lead to better verification performance.

This issue is particularly relevant in practical applications such as voice authentication, access control, and remote communication. In these situations, background noise can reduce the reliability of speaker verification. Although speech enhancement may improve audio clarity, it could also modify speaker-related features and affect authentication accuracy.

Lay and Gerkmann [7] investigated speech enhancement using a speaker verification model as a regularizer, highlighting the importance of preserving speaker-related information during noise reduction. However, classical Wiener filtering does not explicitly optimize speaker identity preservation.

Therefore, this project evaluates how Wiener filtering affects speaker identity under different noise conditions. By comparing signal quality, speaker embeddings, and verification performance, the study examines whether noise reduction provides consistent benefits for speaker verification.

3. Methodology

3.1 Experimental Framework

This project uses an experimental framework combining MATLAB-based speech enhancement and Python-based speaker verification. The overall process includes noise generation, Wiener filtering, speaker embedding extraction, and performance evaluation.

First, clean speech recordings from the LibriSpeech dataset are mixed with environmental noise from the DEMAND dataset at four input SNR levels: −5, 0, 5, and 10 dB. Three noise environments are considered: DKITCHEN, TBUS, and PSTATION.

Next, a fixed STFT-based Wiener filter is applied to the noisy recordings in MATLAB. The clean, noisy, and enhanced speech signals are saved separately for further analysis.

In Python, a pretrained ECAPA-TDNN model extracts speaker embeddings from the processed recordings. Finally, signal quality and speaker verification performance are evaluated using SNR improvement, embedding drift, Mel-filterbank distortion, and Equal Error Rate (EER).

This framework allows the effects of Wiener filtering on signal quality and speaker identity preservation to be examined under consistent experimental conditions.

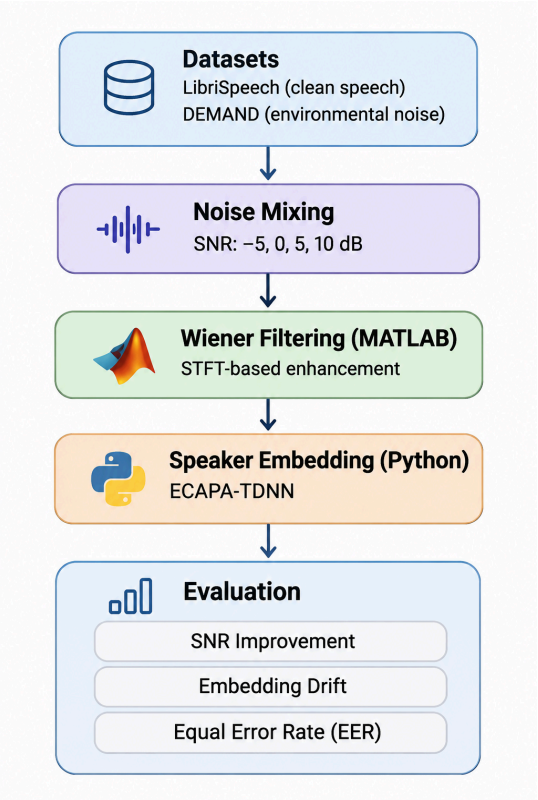


Figure 3.1. Overall experimental framework for Wiener speech enhancement and speaker identity evaluation.

3.2 Speech and Noise Datasets

There's two different datasets used in this project which are LibriSpeech for clean speech recordings and DEMAND for environmental noise.

I selected ten different speakers from the LibriSpeech dev-clean dataset [3].And for each speaker, two different utterances were used. One utterance was used for speaker enrollment and another for verification. This separation allowed speaker identity to be evaluated using different recordings from the same person.

Then I selected three environmental noise recordings from the DEMAND dataset [4],which are DKITCHEN (kitchen noise), TBUS (bus noise), and PSTATION (public station noise). These recordings represent different acoustic environments and provide a range of background noise conditions.

Each verification recording was mixed with the three noise types at four input SNR levels: −5, 0, 5, and 10 dB which makes 120 different experimental conditions (10 speakers × 3 noise types × 4 SNR levels).

Then for each condition, clean, noisy, and Wiener-enhanced speech recordings were saved separately. All recordings were processed at a sampling rate of 16 kHz to ensure compatibility with the speaker verification model.

The following table shows the project dataset clearly

Parameter	Configuration
Clean speech dataset	LibriSpeech dev-clean
Speakers	10
Utterances per speaker	2 (Enrollment and Verification)
Noise dataset	DEMAND
Noise types	DKITCHEN, TBUS, PSTATION
Input SNR	−5, 0, 5, 10 dB
Sampling rate	16 kHz
Experimental conditions	120
Saved audio files	360

Table 3.1. Dataset configuration

3.3 Wiener Filtering Implementation

In this project, wiener filtering was implemented in MATLAB using the Short-Time Fourier Transform (STFT). I divided the noisy speech signal into overlapping frames and transformed it into the frequency domain. Then used a Hann window of 512 samples, a hop size of 256 samples, and a 512-point FFT throughout the experiment.

The noisy speech spectrum was represented as:

$$Y(k,m)=S(k,m)+N(k,m)$$

where (Y(k,m)), (S(k,m)), and (N(k,m)) represent the noisy speech, clean speech, and noise spectra, respectively.

And the Wiener gain was calculated using estimated speech and noise power:

$$G(k,m) = \frac{P_S(k,m)}{P_S(k,m) + P_N(k)}$$

where P_S and P_N are the estimated speech and noise power spectra.The noise power was estimated from a 0.25-second segment of the known DEMAND noise recording and speech power was estimated by subtracting the noise power estimate from the noisy speech power, with negative values set to zero.

The gain was also limited to a minimum value of 0.05 to avoid excessive spectral suppression. Then the enhanced spectrum was reconstructed by using the inverse STFT.

3.4 Speaker Embedding Extraction

After Wiener filtering, Python was used to analyse the clean, noisy, and enhanced speech recordings. A pretrained ECAPA-TDNN model from SpeechBrain was used to extract speaker embeddings [2].

All audio files were processed at 16 kHz. For each recording, the model generated a 192-dimensional embedding vector containing information about the speaker's voice.

Each speaker had two different recordings. One clean recording was used for enrollment, and the other was used for verification. The verification recording was tested in three conditions: clean, noisy, and Wiener-enhanced.

Cosine similarity was used to compare the embedding vectors from the enrollment and verification recordings:

$$C(e_1, e_2) = \frac{e_1^T e_2}{|e_1||e_2|}$$

A higher similarity score means that the two recordings have more similar speaker embeddings.

3.5 Evaluation Metrics

After processing the speech recordings, I used four different measurements to compare the clean, noisy, and Wiener-enhanced speech which are SNR improvement, embedding drift, Mel-filterbank distortion, and Equal Error Rate (EER).

First, SNR improvement was calculated to measure the effect of noise reduction:

$$\Delta SNR = SNR_{enhanced} - SNR_{noisy}$$

Positive result means that Wiener filtering improved the signal-to-noise ratio.

Then embedding drift was used to measure how much the speaker embedding changed compared with the original clean speech:

$$D = 1 - C(e_{clean}, e_{processed})$$

A smaller value means that the processed embedding is closer to the clean speech embedding. Embedding recovery was also calculated by subtracting the Wiener drift from the noisy drift.

Mel-filterbank distortion was measured using 80 Mel-frequency bands to compare the spectral features of the processed speech with the clean speech. A lower value means less difference between their features.

Finally, EER was calculated using the cosine similarity scores from speaker verification. Each condition included 10 same-speaker comparisons and 90 different-speaker comparisons. A lower EER indicates better verification performance.

4. Results and Discussion

4.1 SNR Improvement

The first experiment measured the SNR improvement after Wiener filtering. The results were calculated using the saved noisy and enhanced speech recordings under three noise environments and four input SNR levels.

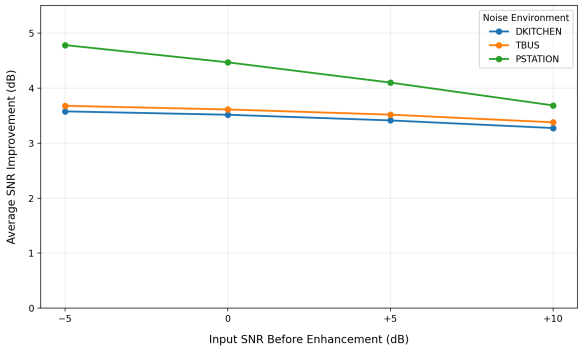


Figure 4.1. Average SNR improvement after Wiener filtering under three noise environments and four input SNR levels.

As shown in Figure 4.1, Wiener filtering improved the SNR in all 12 tested conditions. The average SNR improvement across 120 recordings was approximately 3.75 dB.

Among these three noise environments, PSTATION showed the largest improvement. At −5 dB input SNR, the improvement reached 4.78 dB. DKITCHEN and TBUS also showed positive results, with improvements ranging from approximately 3.27 to 3.68 dB.

The results show that Wiener filtering reduced background noise across the tested conditions.

4.2 Embedding Drift and Spectral Distortion

The second experiment mainly focused on how Wiener filtering affected speaker embeddings and spectral features. Embedding drift was used to measure the difference between processed speech and the corresponding clean speech, while log Mel-filterbank distortion was used to compare their spectral features.

Evaluation Metric	Noisy Speech	Wiener-enhanced Speech
Average Embedding Drift	0.2114	0.2371
Average Log Mel-filterbank Distortion	3.2406	2.2931

Table 4.2. Average speaker embedding drift and log Mel-filterbank distortion before and after Wiener filtering.

As shown in Table 4.2: After Wiener filtering, the average embedding drift increased from 0.2114 for noisy speech to 0.2371. The average embedding recovery was $R_{emb} = 0.2114 - 0.2371 = -0.0257$.

However, the spectral results showed a different trend. The average log Mel-filterbank distortion decreased from 3.2406 to 2.2931, representing a reduction of approximately 29.2%.

These results show that Wiener filtering improved spectral similarity to clean speech, but did not consistently preserve speaker embedding similarity.

4.3 Speaker Verification Performance

The third experiment evaluated speaker verification performance using Equal Error Rate (EER). The results compared clean, noisy, and Wiener-enhanced speech under different noise conditions. A lower EER indicates better speaker verification performance.

Noise Type	Input SNR (dB)	Noisy EER	Wiener EER
DKITCHEN	−5	10.00	10.00
DKITCHEN	0	10.00	8.33

DKITCHEN	+5	8.33	1.67
DKITCHEN	+10	2.22	1.11
TBUS	−5	2.22	2.22
TBUS	0	1.67	1.67
TBUS	+5	1.67	1.11
TBUS	+10	1.11	0.56
PSTATION	−5	2.22	10.56
PSTATION	0	0.00	0.56
PSTATION	+5	0.00	0.00
PSTATION	+10	0.00	0.00
Average	——	3.29	3.15

As shown in Table 4.3, the average EER decreased from 3.29% to 3.15% after Wiener filtering. However, the improvement was not consistent across all conditions.

For example, under DKITCHEN noise at +5 dB input SNR, the EER decreased from 8.33% to 1.67%. In contrast, under PSTATION noise at −5 dB, the EER increased from 2.22% to 10.56% after enhancement.

In conclusion, Wiener filtering improved speaker verification performance in some conditions but reduced it in others. These results show that improving speech SNR does not always lead to better speaker verification performance.

4.4 Discussion and Limitations

The results show that Wiener filtering improved speech quality but did not always preserve speaker identity information. The average SNR improvement was about 3.75 dB, and the average log Mel-filterbank distortion decreased by 29.2%. These results show that Wiener filtering reduced noise and improved the spectral similarity between enhanced and clean speech.

However, the speaker embedding results showed a different trend. The average embedding drift increased from 0.2114 to 0.2371 after enhancement, with an average embedding recovery of −0.0257. This suggests that Wiener filtering may change some acoustic features used by the ECAPA-TDNN model to represent speaker identity. Although the enhanced speech became spectrally closer to clean speech, its speaker embeddings did not always become more similar.

The speaker verification results further demonstrated this difference. The average EER decreased from 3.29% to 3.15%. However, performance varied across noise conditions. For example, under PSTATION noise at −5 dB, the EER increased from 2.22% to 10.56% after enhancement. This shows that better noise reduction does not relate to better results of speaker verification.

There are still several limitations that should be considered in this project.

First, only 10 speakers are used from LibriSpeech, which limits the reliability of the results.

Second, only three noise environments and four SNR levels were evaluated, so the results may not represent all real-world conditions.

Finally, the evaluation used one pretrained ECAPA-TDNN model, and other speaker verification models may respond differently to the same enhancement method.

Future work could include more speakers, additional noise environments, and different enhancement methods. It would also be useful to investigate noise estimation without access to a reference noise signal and to explore enhancement methods that better preserve speaker-related acoustic features.

5. Conclusion

This project investigated the effect of Wiener speech enhancement on speaker identity preservation under different noise conditions. MATLAB was used for speech enhancement, while a pretrained ECAPA-TDNN model was used to evaluate speaker identity information and verification performance.

The results demonstrated that Wiener filtering can improve speech quality, but better noise reduction does not always lead to better speaker verification performance. This highlights the importance of evaluating both speech quality and speaker identity preservation when developing speech enhancement systems.

References

[1] J. S. Lim and A. V. Oppenheim, “Enhancement and Bandwidth Compression of Noisy Speech,” Proceedings of the IEEE, vol. 67, no. 12, pp. 1586–1604, 1979. DOI: 10.1109/PROC.1979.11540.

[2] B. Desplanques, J. Thienpondt, and K. Demuynck, “ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification,” Interspeech 2020, pp. 3830–3834, 2020. DOI: 10.21437/Interspeech.2020-2650.

[3] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “LibriSpeech: An ASR Corpus Based on Public Domain Audio Books,” ICASSP, pp. 5206–5210, 2015. DOI: 10.1109/ICASSP.2015.7178964.

[4] J. Thiemann, N. Ito, and E. Vincent, “DEMAND: A Collection of Multi-channel Recordings of Acoustic Noise in Diverse Environments,” Zenodo, 2013. DOI: 10.5281/zenodo.1227120.

[5] Y. Ephraim and D. Malah, “Speech Enhancement Using a Minimum Mean-Square Error Short-Time Spectral Amplitude Estimator,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 32, no. 6, pp. 1109–1121, 1984, doi: 10.1109/TASSP.1984.1164453.

[6] D. Snyder, D. Garcia-Romero, G. Sell, D. Povey, and S. Khudanpur, “X-Vectors: Robust DNN Embeddings for Speaker Recognition,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2018, pp. 5329–5333, doi: 10.1109/ICASSP.2018.8461375.

[7] B. Lay and T. Gerkmann, “Speech Enhancement Regularized by a Speaker Verification Model,” in *Proc. IEEE 24th International Workshop on Multimedia Signal Processing (MMSP)*, 2022, doi: 10.1109/MMSP55362.2022.9949479.